Alvy Villatoro

HW #9

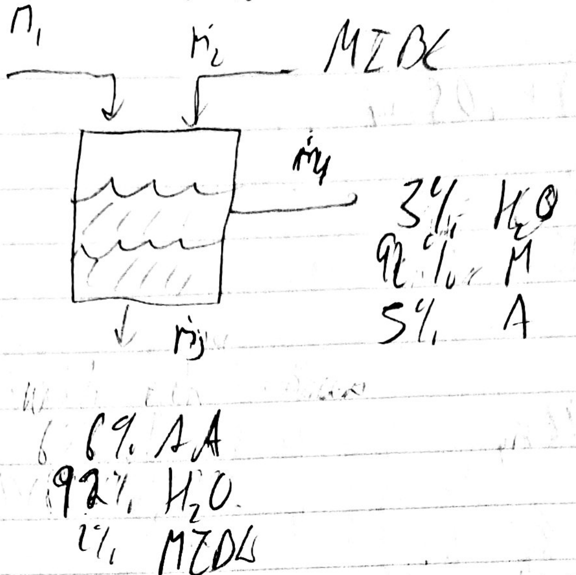
1)

1) 20

2) 10

3) 15 $\frac{60}{100}$

4) 15



Assume $\dot m_1 = 100$ kg/hr

AA $\quad 25 = .06\dot m_3 + .05\dot m_4$ $\quad$ •3 unk $\quad$ 3 eqns

$H_2O \quad 75 = .92\dot m_3 + .03\dot m_4$

MIBK $\quad \dot m_2 = .02\dot m_3 + .92\dot m_4$

$$\frac{25 - .06\dot m_3}{.05} = \dot m_4$$

$$75 = .92\dot m_3 + .03\left(\frac{25 - .06\dot m_3}{.05}\right)$$

$$\left(75 = .92\dot m_3 + \frac{.75 - .0018\dot m_3}{.05}\right) .05$$

$$3.75 = .046\dot m_3 + .75 - .0018\dot m_3$$

$$3.00 = .046\dot m_3 - .0018\dot m_3$$

$$\frac{3.00}{.0442} = \frac{.0442\dot m_3}{.0442} \qquad \dot m_3 = 67.87 \frac{kg}{hr}$$

$25 = .06(67.87) + .05\dot{m}_4$

$\dot{m}_4 = 418.55 \frac{kg}{hr}$ ✓

$\dot{m}_L = .02\dot{m}_3 + .92\dot{m}_4$

$\dot{m}_L = 386.42 \frac{kg}{hr}$ ✓

$\frac{m\ \text{Solvent}}{m\ \text{Feed}} = ?$

$\frac{20}{25}$

1L of Water   1g of phenol   5g activated Carbon at 20°C

adsorbed phenol $\frac{\text{mmol}}{\text{g of C}}$ phenol soln $\frac{\text{mmol}}{\text{L of } H_2O}$

10.638 = Phenol conc (1L) + adsorbed phenol (5g) → Trial an Error → Removal

1g phenol → $\frac{1 \text{mol}}{94.1\text{g}}$ → .010625863 mol → $\frac{1000 \text{ mmol}}{1 \text{ mol}}$ → 10.6258(335 mmol phenol

$\frac{10.62586335 \text{ mmol phenol}}{1L \text{ of } H_2O}$ $\frac{10.62586335 \text{ mmol phenol}}{5g\ C}$

$\frac{10}{25}$ 2.125 → ? what is this?

How much is Removed? (adsorbed ph. x 5g)

3) 25°C → 150°C · $\Delta H' = 3640$ J/mol

$V = 1.25 \frac{m^3}{\text{min}}$ $P = 122$ kPa → $\frac{1 \text{atm}}{101.325}$ → 1.204 atm

$\Delta H + \cancel{\Delta E_k} + \cancel{\Delta E_p} = Q + \cancel{W}$ no movement no moving parts

$\Delta H = \dot{Q}$ $PV = \dot{n}RT$ $\frac{PV}{RT} = \dot{n}$ $\dot{n} = .04334 3839 \frac{\text{mols}}{\text{min}}$

43.3 $\frac{\text{mol}}{\text{min}}$ $\frac{\cancel{15}}{25}$

$\dot{n}\Delta H = \dot{Q}$

$\dot{Q} = .04334 3839 \frac{\text{mols}}{\text{min}} \cdot 3640 \frac{\text{J}}{\text{mol}} \rightarrow \frac{1 \text{KJ}}{1000 \text{J}}$

this is min

$\frac{\cancel{25}}{\cancel{25}}$

$\dot{Q} = .158 \frac{\text{KJ}}{\text{min}} \cdot \frac{10 \text{ min}}{1 \text{hr}} = 9.466$ kW

watt = J/S   2.63

1) ΔE_k would be positive because of the temperature increase. As the temperature increases the velocity will increase.

more information is needed

4) 1gmol of H_2O $\quad 1mol \rightarrow \frac{18.02g}{1mol} = 18.02g\,W \rightarrow \frac{1kg}{1000g}$ .01802 kg

a) $\Delta E_k = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 =$ ~~90.1~~ kJ 6.9

b) $\Delta E_p = mg\Delta h = mg(h_f - h_i) = mgh_f = 17.66$ kJ

c)

2

d) $\Delta H = mC_p\Delta T = mC_p(373.15 - 273.15) = 6.79$ kJ 7540

15/25

e) $H_2O_{(l)} \rightarrow H_{2(g)} + O_{2(g)}$

284600 J

2 $0 + 0 - (-292.740) = 292.740$ kJ ~~288000~~

10^{15}

f) $E = mC^2 = 1.6218 \cdot 10^{14}$ kJ

please include your complete calculations next time!